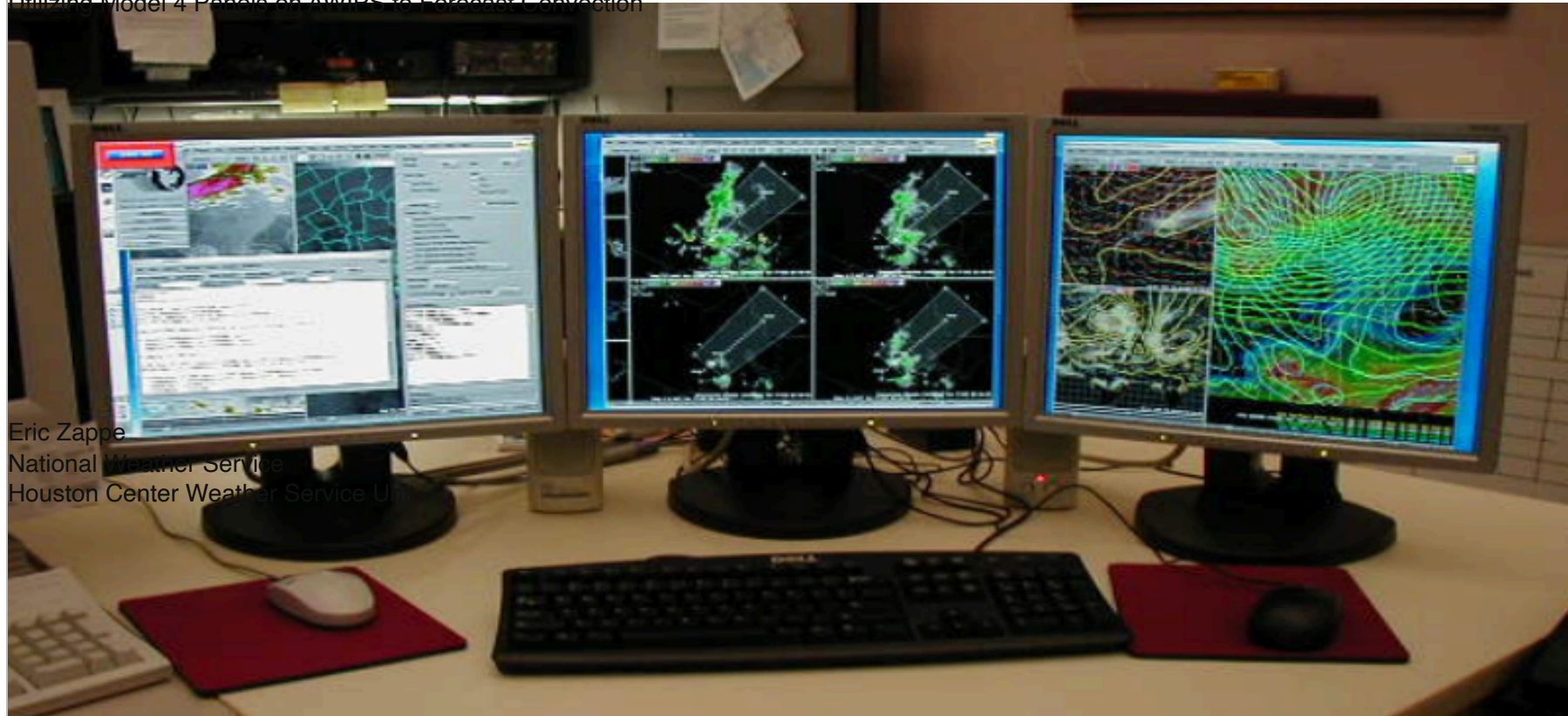


## Utilizing Model 4 Panels on AWIPS to Forecast Convection



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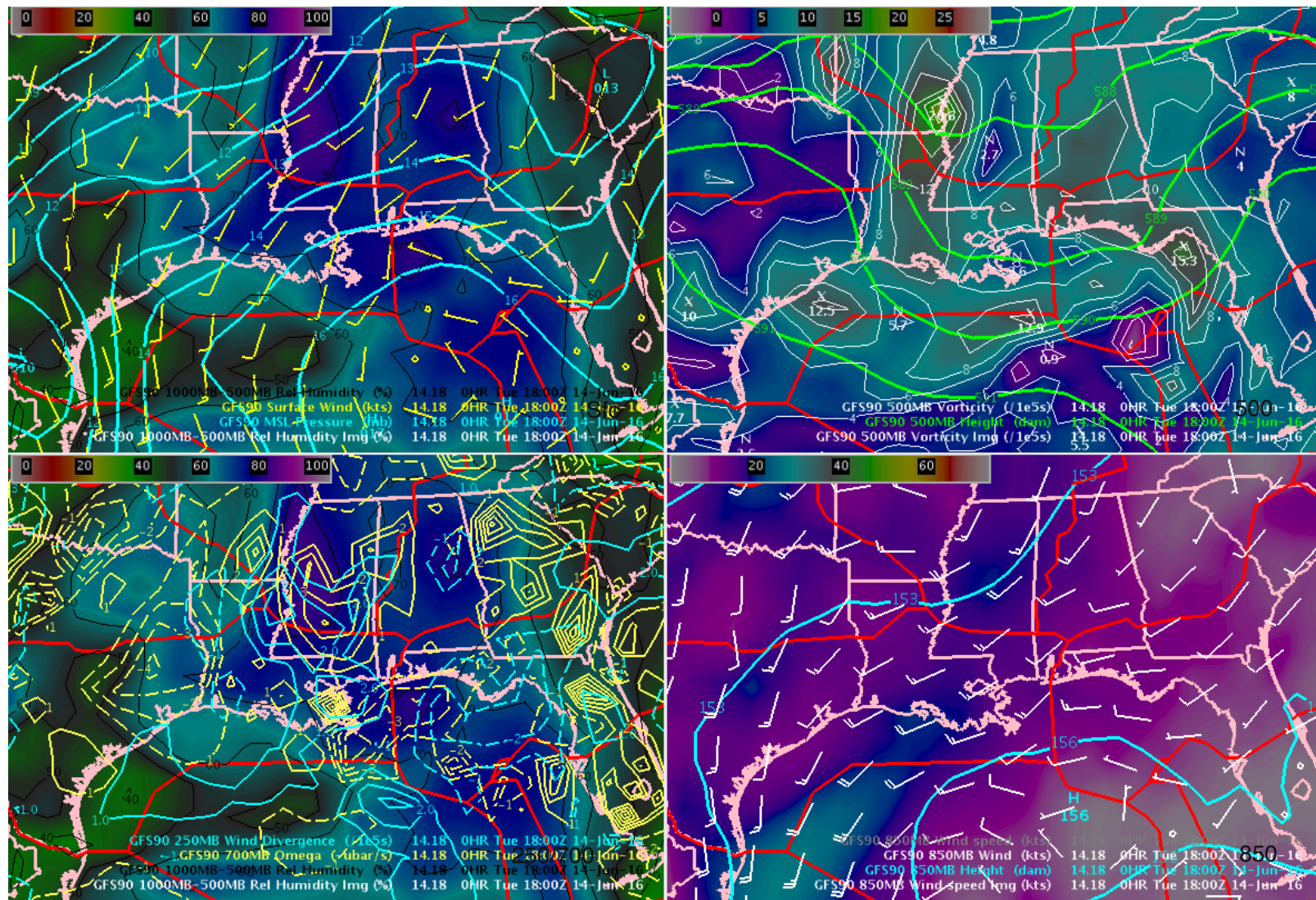


This tutorial will demonstrate the usefulness of employing model 4 panels to forecast timing/place of convection.

This event took place during the afternoon of June 14, 2016 across Houston CWSU (ZHU) portions of the Gulf Coast region.

The following case study contains AWIPS 4-panel snapshots of GFS and RAP forecast models.

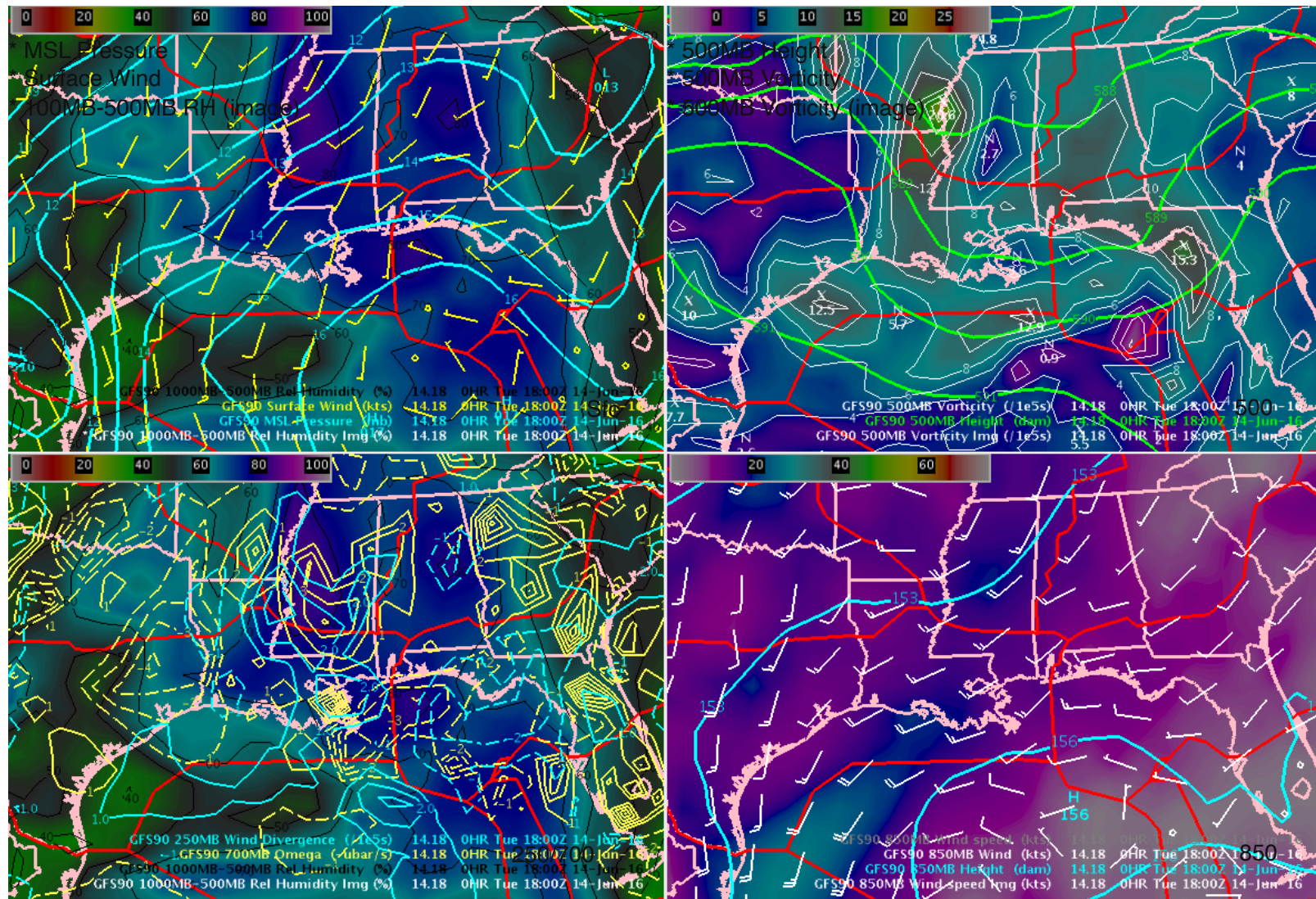
I first like to view GFS model to discern large (synoptic) scale features...





Top left is surface features

Top right is 500 mb features

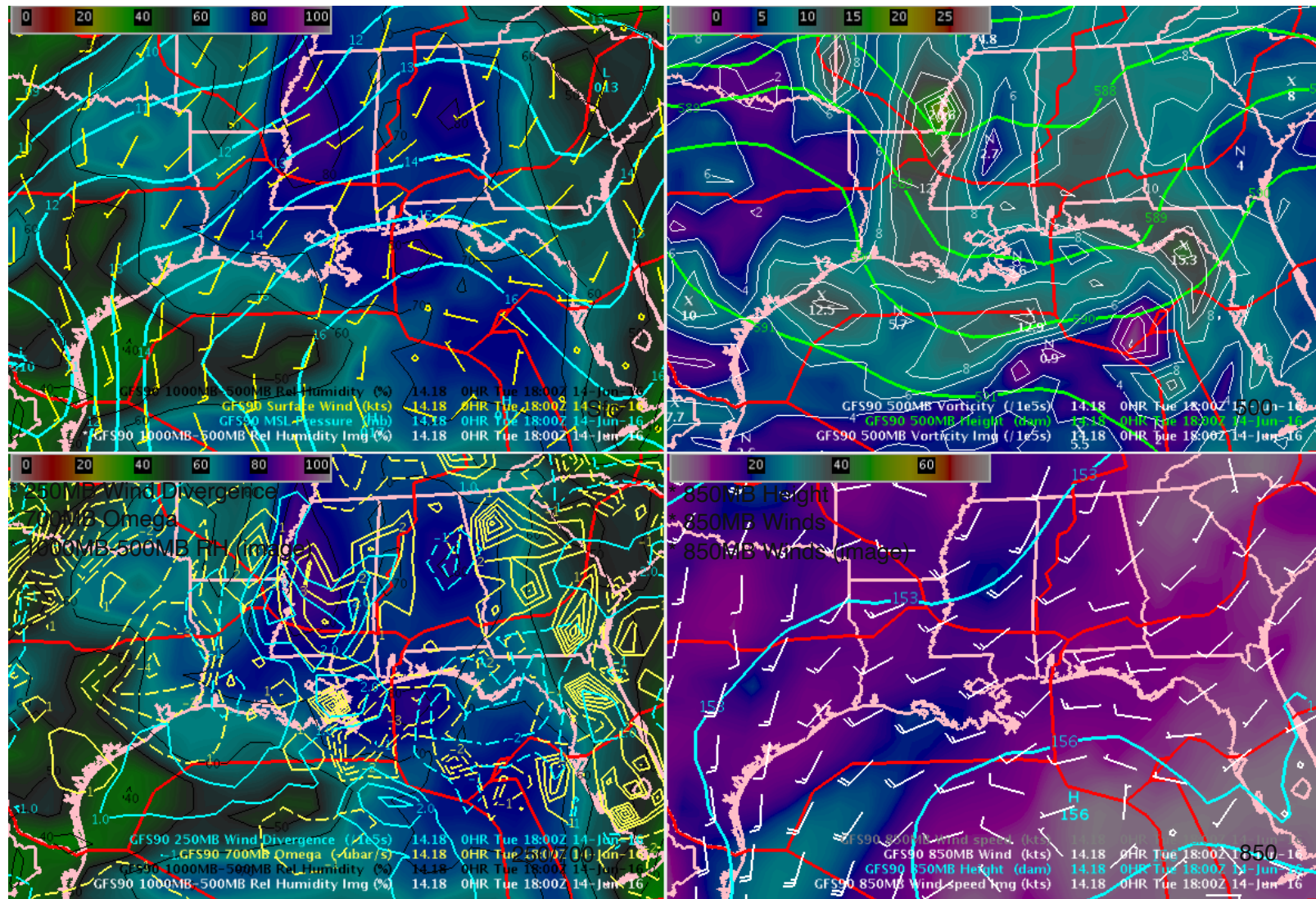


Slide 4

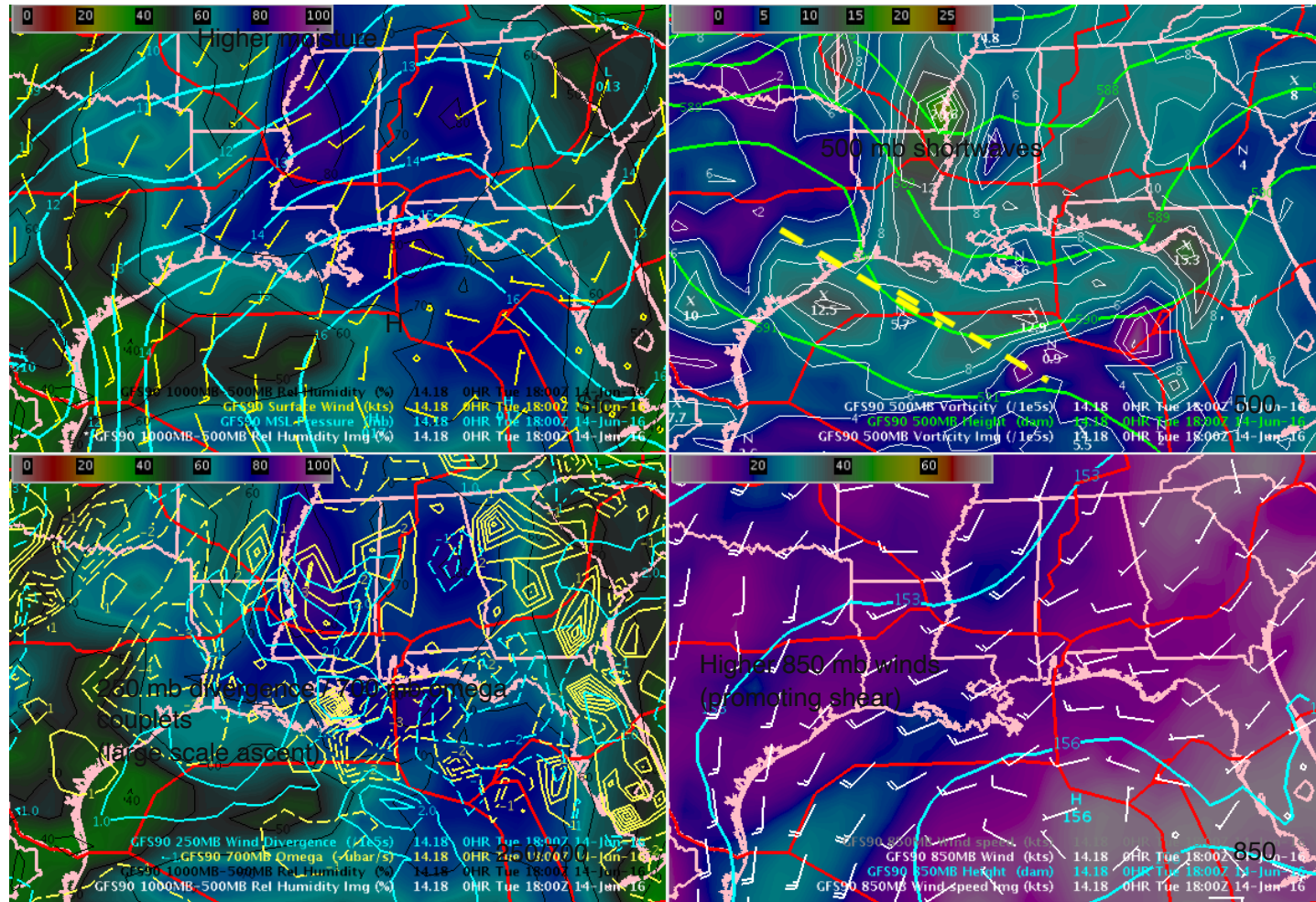


Bottom left is 700/250 mb features

Bottom right is 850 mb features

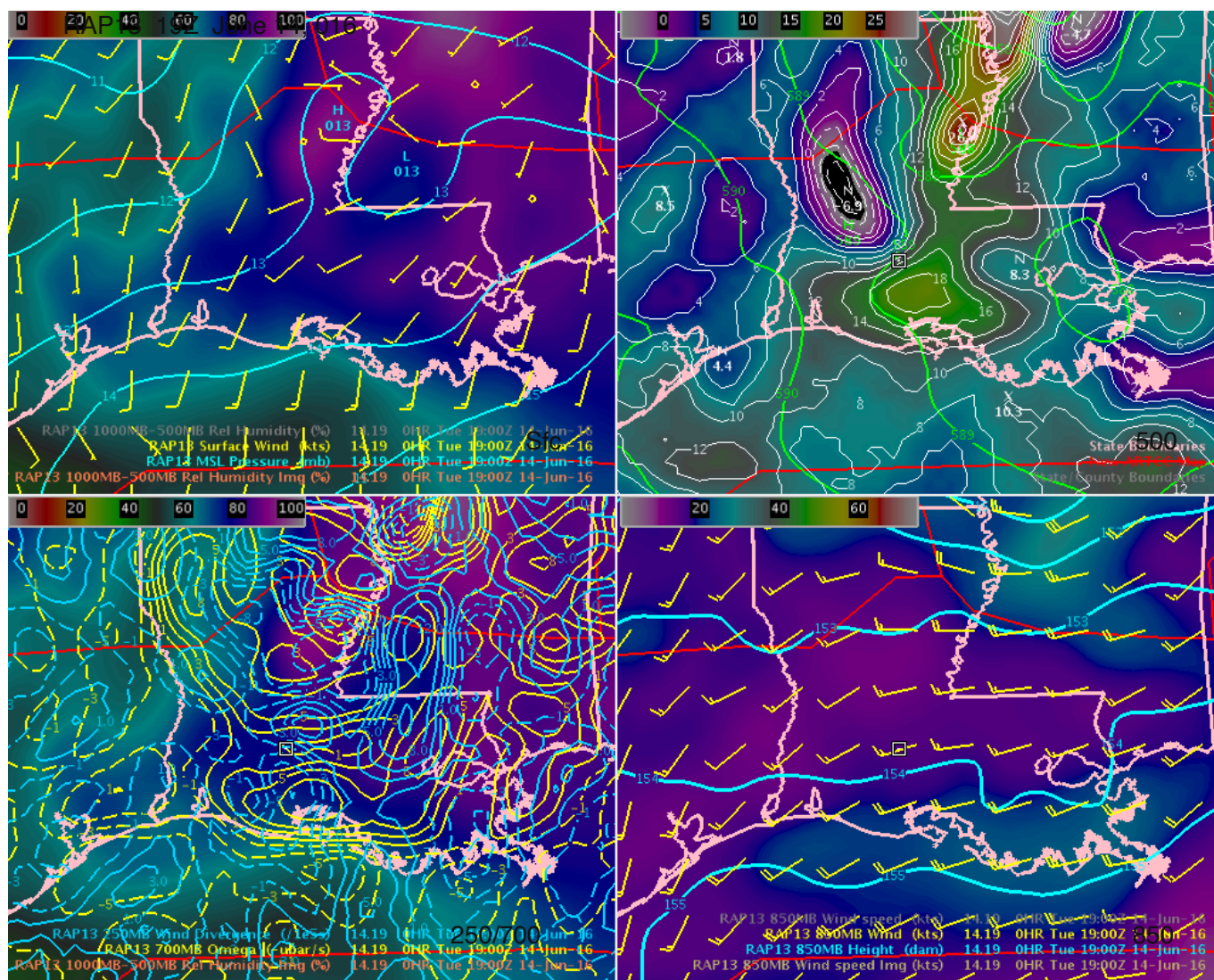


A broad trough (and associated lift) was noted from Louisiana eastward as well as over the northern half of the Gulf of Mexico



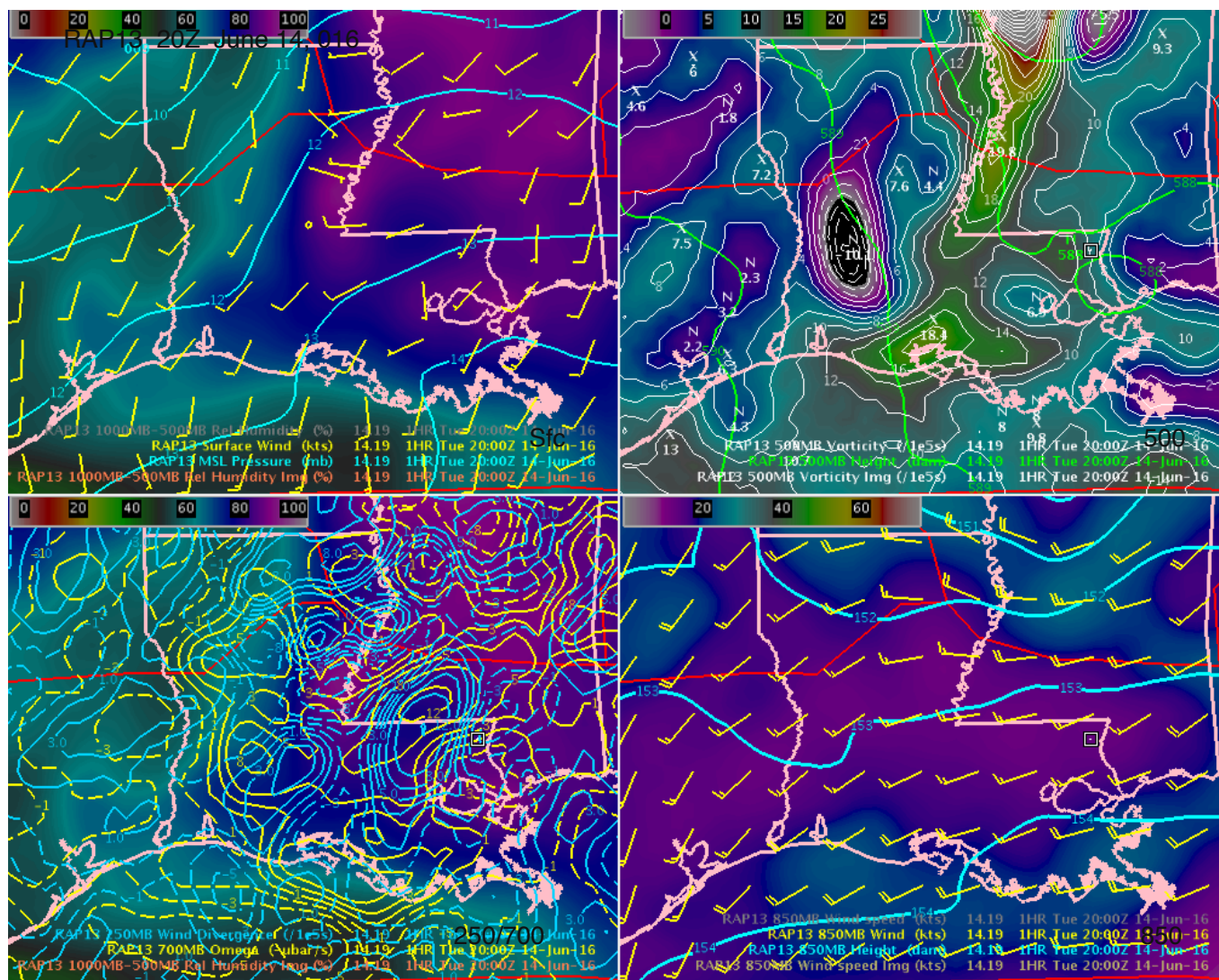


then look at the finer details using same 4-panel configuration of 13 km RAP model

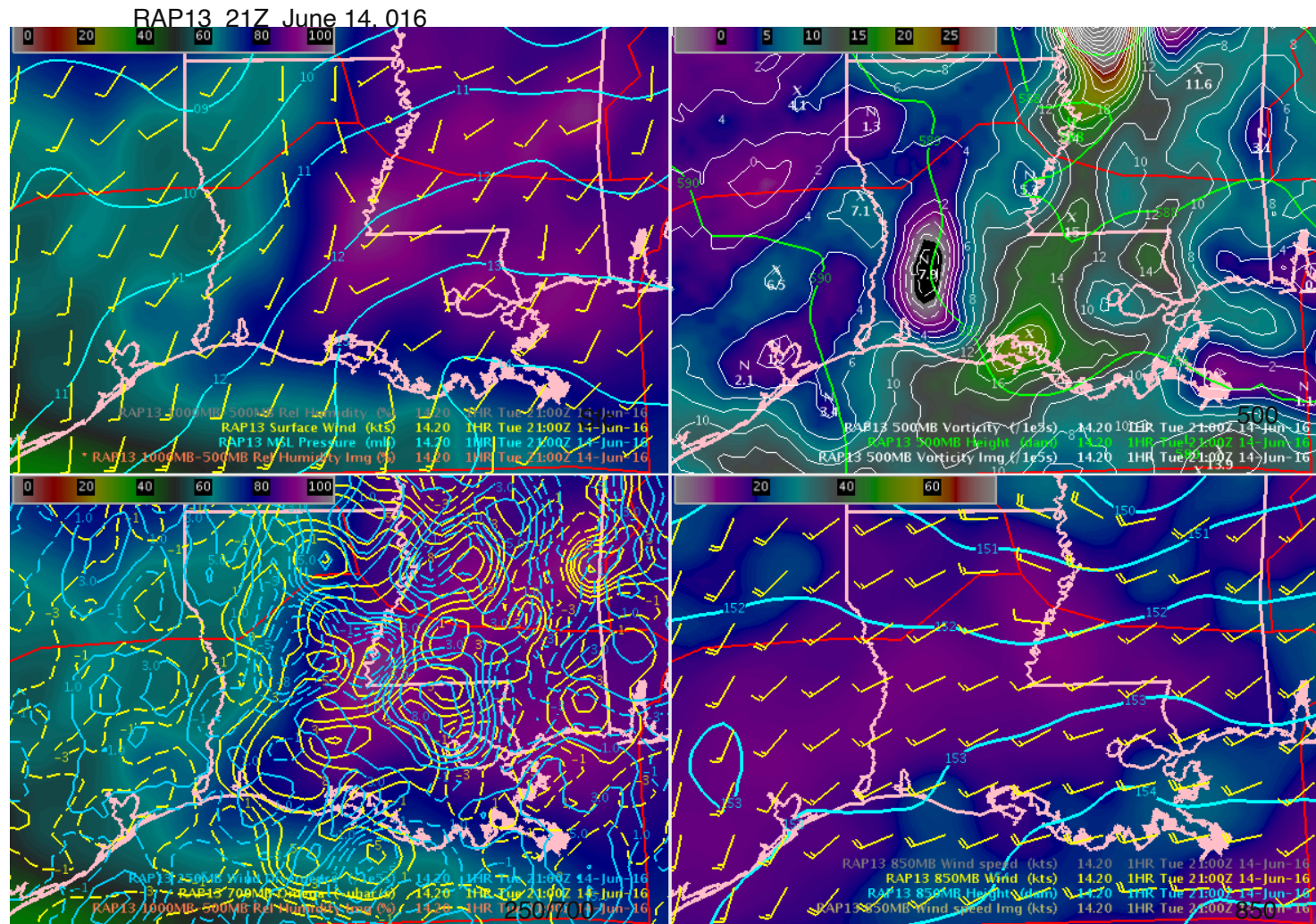




RAP13 provides one-hour time steps, so I advance to see trends of weather features...

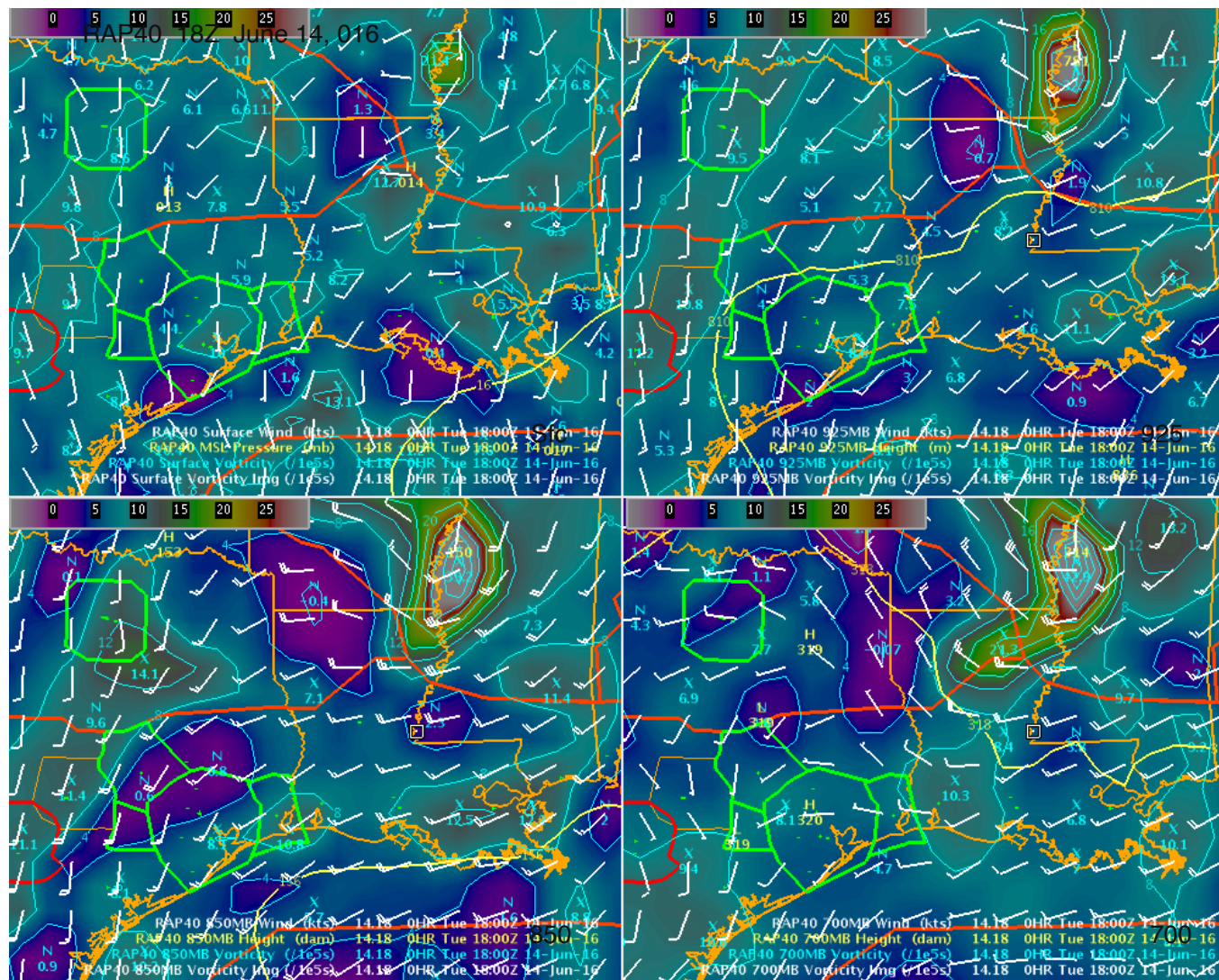


such as strengthening/weakening of winds and 500 mb shortwaves

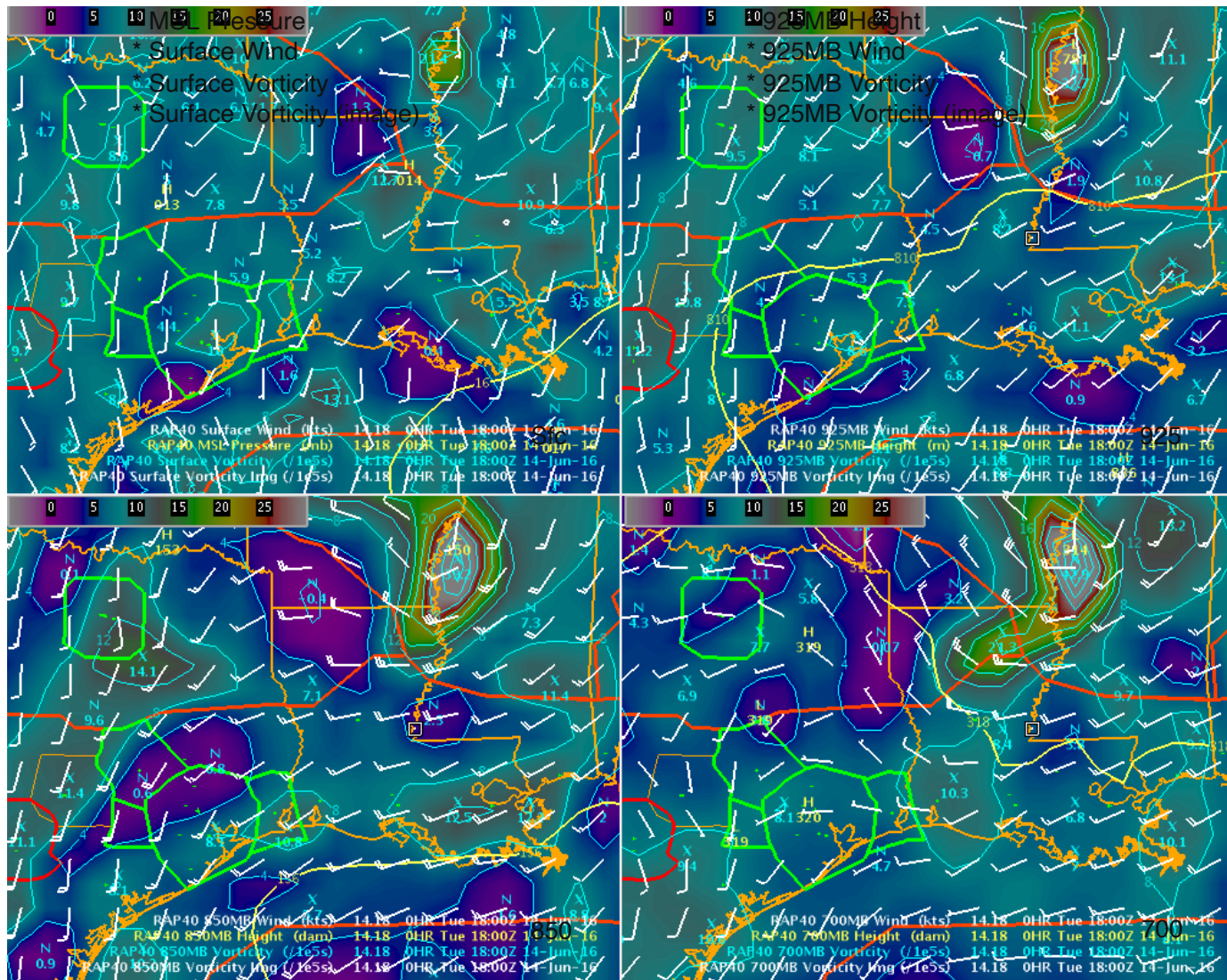




I also like to see lowest slices of the atmosphere to identify lifting mechanisms

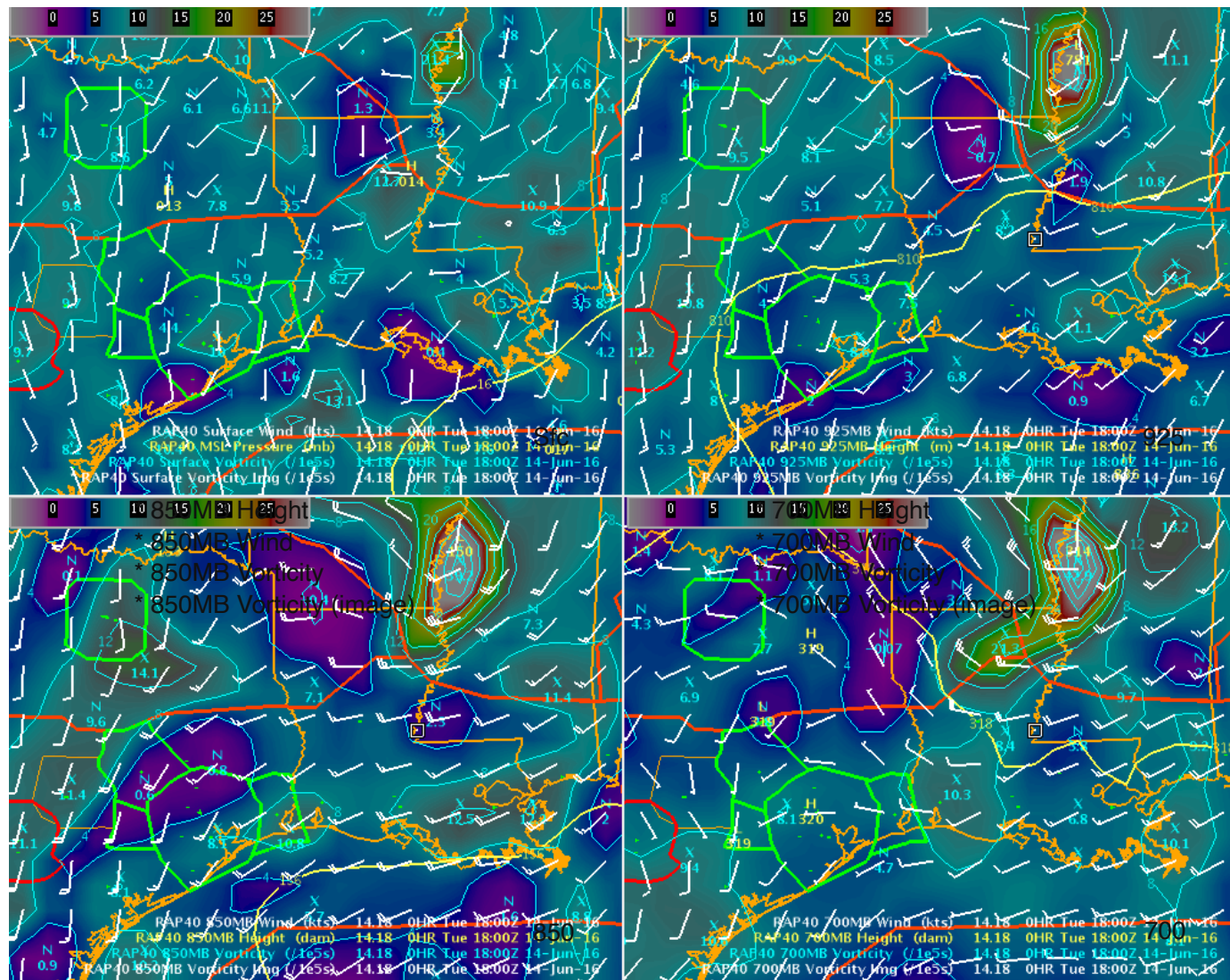


Top left is surface features  
Top right is 925 mb features

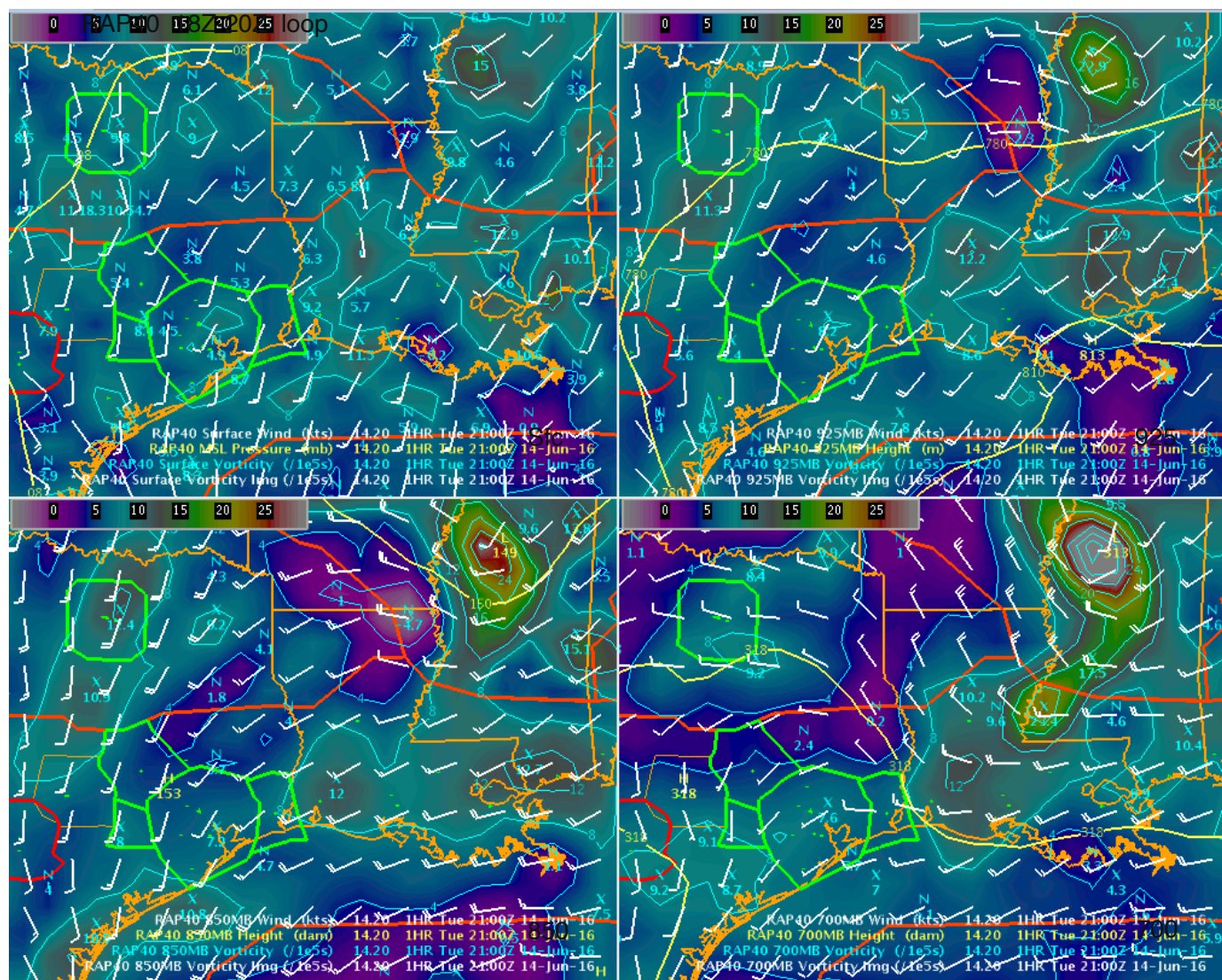




Bottom left is 850 mb features  
Bottom right is 700 mb features

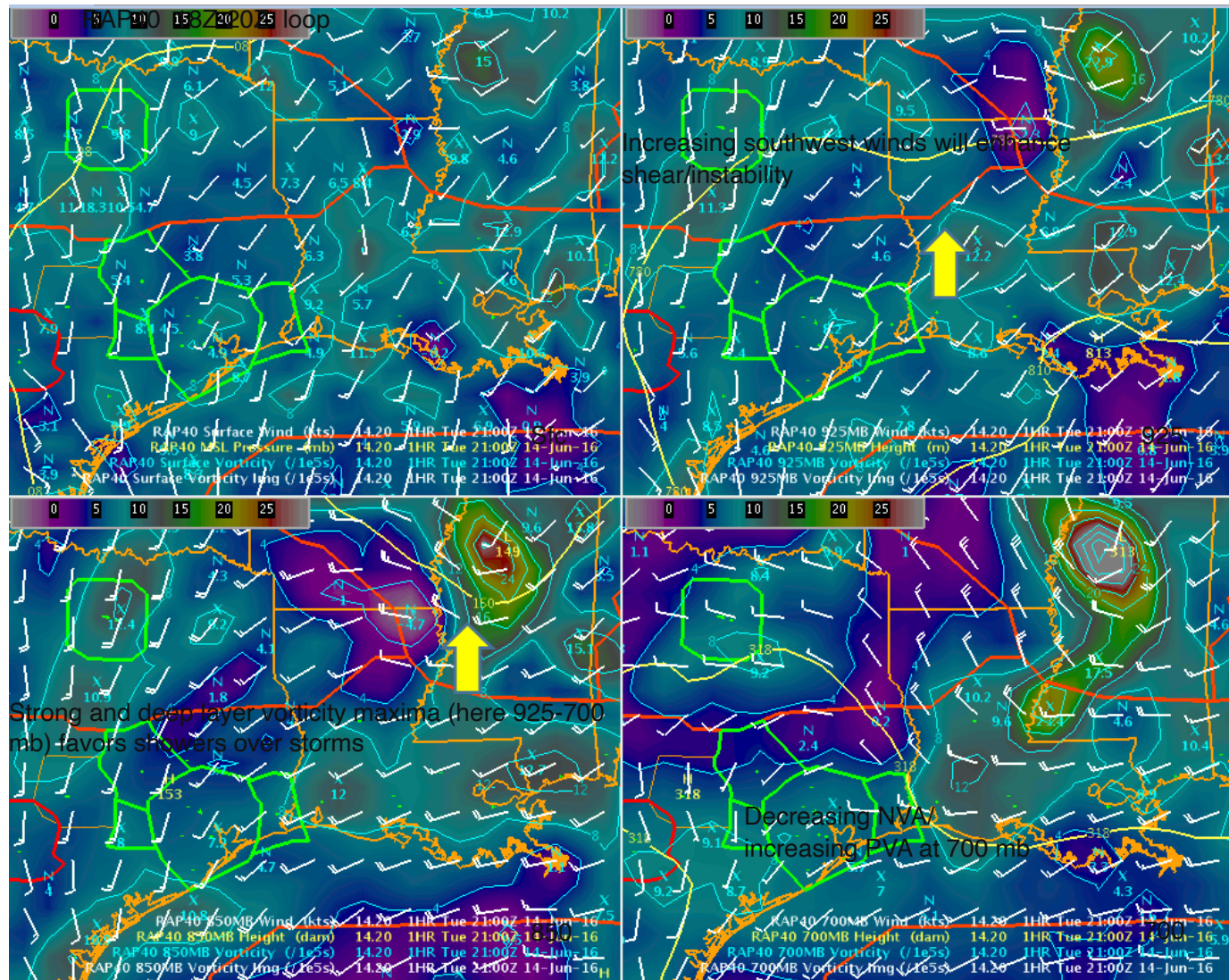


I like to loop these images to detect trends, such as increasing/decreasing PVA/NVA

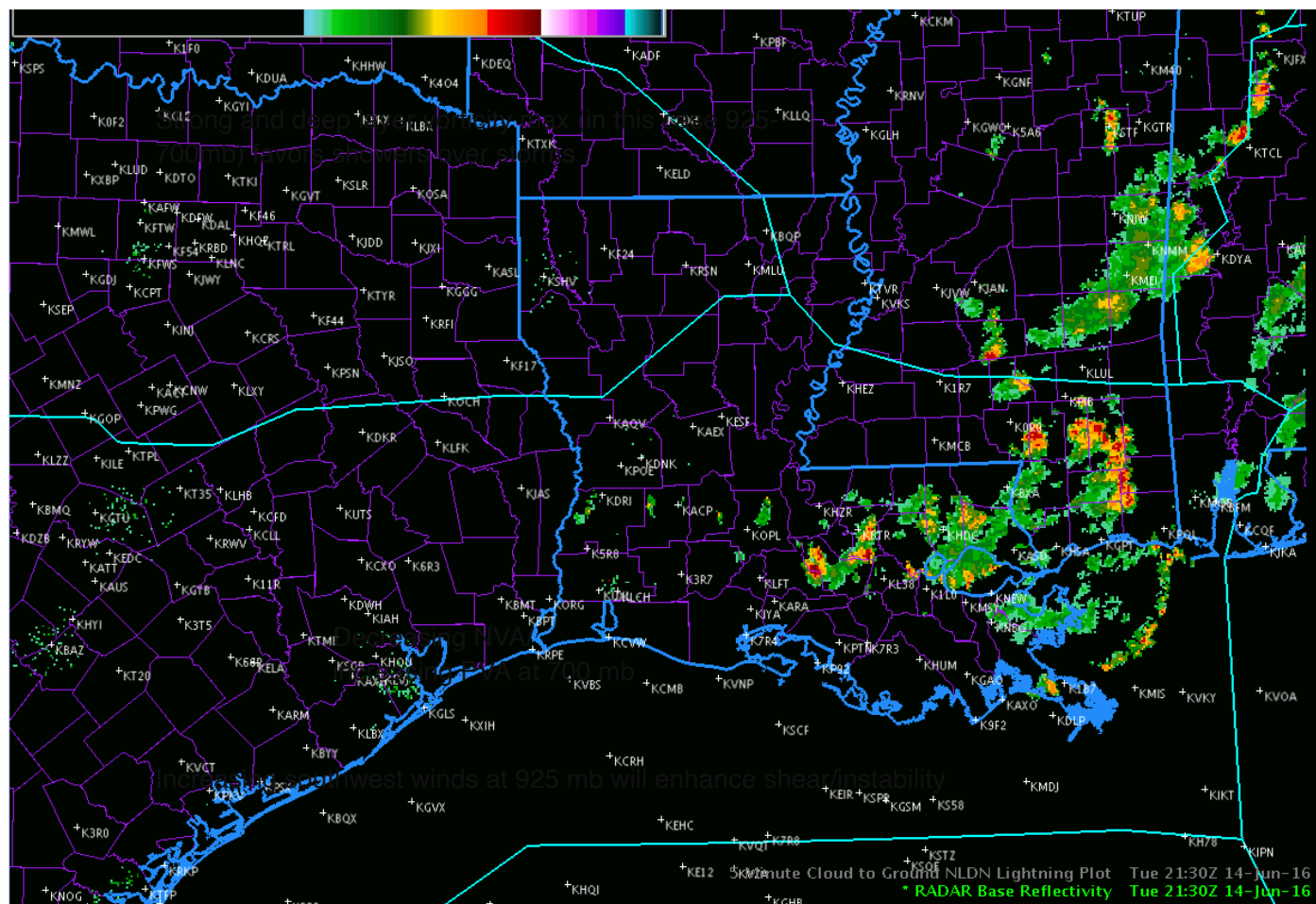




I like to loop these images to detect trends, such as increasing/decreasing PVA/NVA



Here's the mosaic radar loop for the western half of the Gulf Coast region

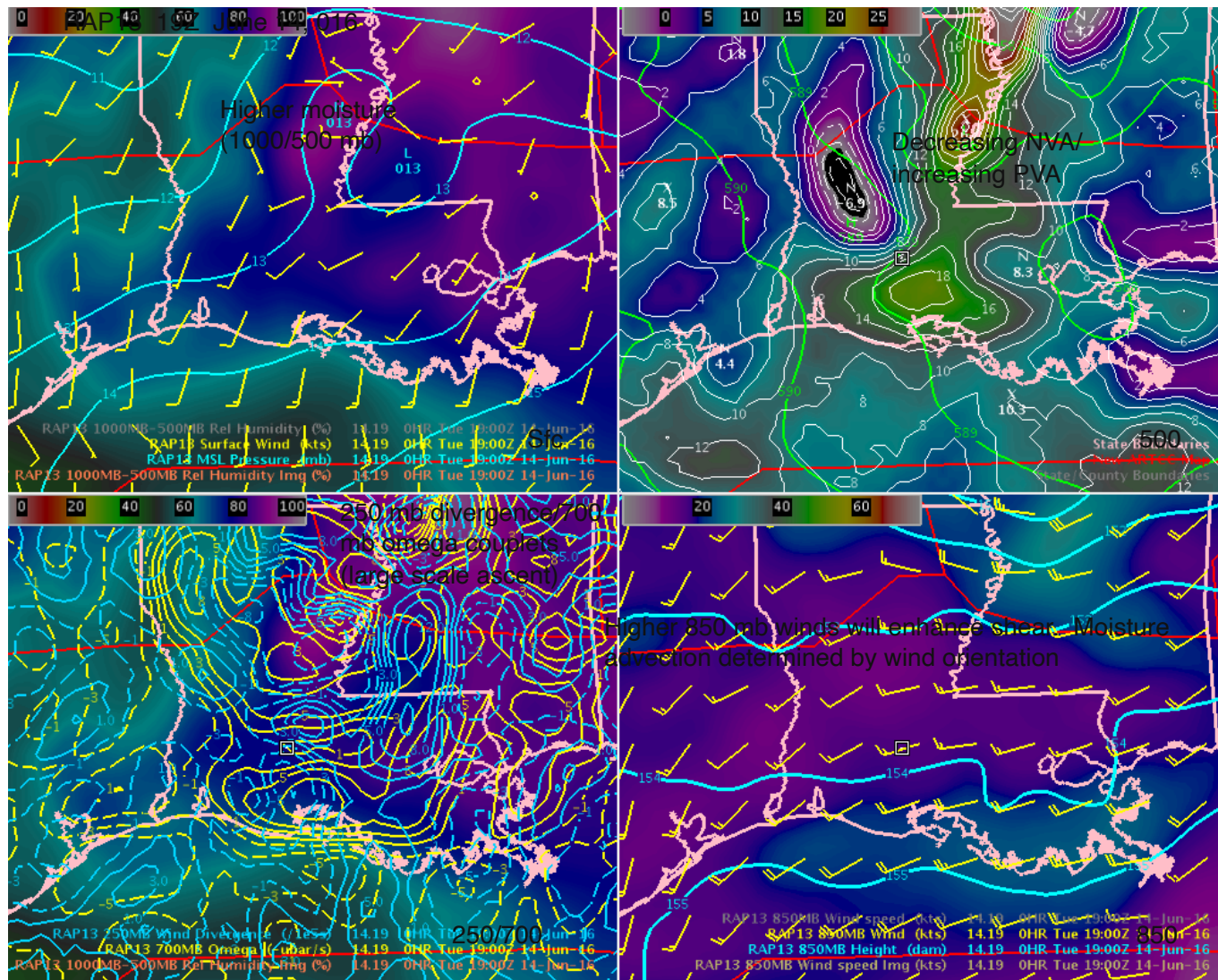




5-minute strikes green      15-minute (older) strikes red



Once again, the other 4 panel of RAP model





Now for quick review of necessary ingredients for thunderstorm development

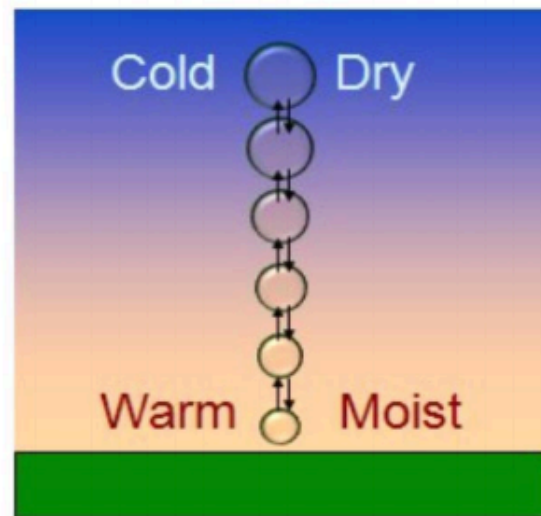


# Necessary Ingredients for Thunderstorm Cell Formation

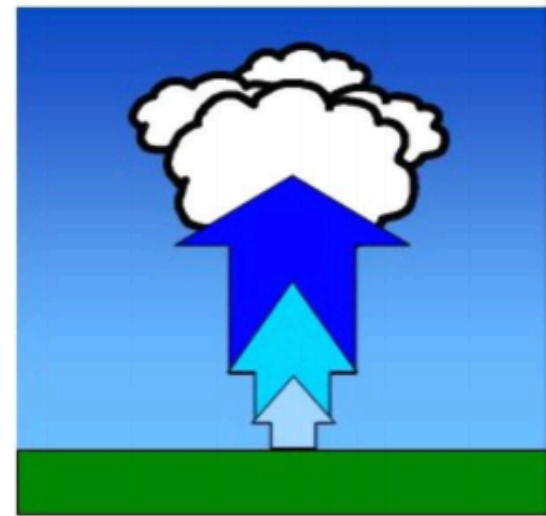
Water Vapor



Unstable Air



Lift



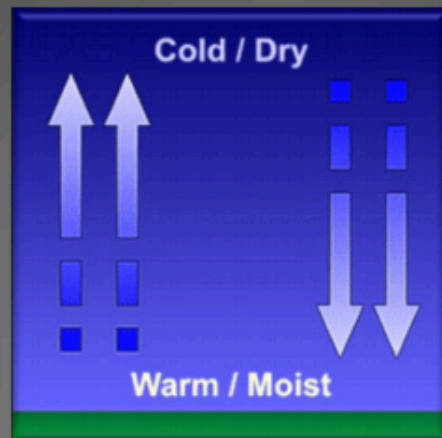
# Thunderstorm Ingredient - Moisture



Typical sources of moisture are large bodies of water such as the Atlantic and Pacific oceans as well as the Gulf of Mexico. Warm ocean currents occur along east coasts of continents with cool ocean currents occur along west coasts. Evaporation is higher in warm ocean currents and therefore put more moisture into the atmosphere than with cold ocean currents at the same latitude.



# Thunderstorm Ingredient - Instability



Air is considered unstable if it continues to rise when given a nudge upward (or continues to sink if given a nudge downward). An unstable air mass is characterized by warm moist air near the surface and cold dry air aloft.

In these situations, if a bubble or parcel of air is forced upward it will continue to rise on its own. As this parcel rises it cools and some of the water vapor will condense forming the familiar tall cumulonimbus cloud that is the thunderstorm.

# Thunderstorm Ingredient – Lift



Typically, for a thunderstorm to develop, there needs to be a mechanism which initiates the upward motion, something that will give the air a nudge upward. This upward nudge is a direct result of air density.

Some of the sun's heating of the earth's surface is transferred to the air which, in turn, creates different air densities. The propensity for air to rise increases with decreasing density. This difference in air density is the main source for lift and is accomplished by several methods.

(continued)

# Thunderstorm Ingredient – Lift

## Differential Heating

The sun's heating of the earth's surface is not uniform. For example, a grassy field will heat at a slower rate than a paved street. A body of water will heat slower than the nearby landmass. This will create two adjacent areas where the air is of different densities. The cooler air sinks, pulled toward the surface by gravity, forcing up the warmer, less dense air, creating thermals.

## Fronts, Drylines and Outflow Boundaries

**Fronts** are the boundary between two air masses of different temperatures and therefore different air densities. The colder, more dense air behind the front lift warmer, less dense air abruptly. If the air is moist thunderstorms will often form along the cold front.

(continued)



# Thunderstorm Ingredient – Lift

**Drylines** are the boundary between two air masses of different moisture content and separates warm, moist air from hot, dry air. Moist air is less dense than dry air. Drylines therefore, act similar to fronts in that a boundary exists between the two air masses of different densities. The air temperature behind a dryline is often much higher due to the lack of moisture. But the moist air ahead of the dryline has an even lower density making it more buoyant. The end result is air lifted along the dryline forming thunderstorms.

**Outflow boundaries** are a result of the rush of cold air as a thunderstorm moves overhead. The rain-cooled, more dense, air acts as a "mini cold front", called an outflow boundary. Like fronts, this boundary lifts warm moist air and can cause new thunderstorms to form.

The End

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